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Enhancing Problem-Solving Skills with AI: A Case Study on Innovation and Creativity in a Business Setting

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Abstract

The adoption of artificial intelligence has risen, yet research on its impact on innovation processes between actual businesses remains sparse. This research fills the present gap by investigating ten workers from a tech startup who utilize artificial intelligence tools in operational and creative activities. The paper analyzes business-related AI functionality through a qualitative analysis of ten tech start-up employees. The examination reveals that AI produces significant enhancements in problem resolution by executing mundane actions while analyzing large datasets to deliver data-driven suggestions to users. The interview respondents mentioned that AI's role in diminishing supply chains is 15%, while allowing AI to manage customer service without employee engagement in 80% of interactions. The implementation costs, along with data dependency and occasional contextual blindness in AI systems, represented some of the problems in this system. Analysis demonstrated that AI tools enable the development of innovative concepts and challenge established viewpoints, prompting participants to create a gamified loyalty system and dynamic content planning. Participants in the study emphasized the need for human involvement to refine AI-based insights, recognizing how human imagination complements AI capabilities effectively. The work enhances academic discussions about AI-related problem-solving and creativity while offering specific business-related recommendations for implementation. The recommendations begin with establishing initial experimental programs, while providing support for employee's skills development, and fostering strong alliances between technical AI personnel and professional subject matter experts. Research topics focused on AI application fields and the anticipated impacts on company decision-making, as well as the ethical ramifications, need further exploration. This research confirms the revolutionary potential of artificial intelligence systems for problem-solving methods, but requires proper execution, along with human supervision, to fully realize their advantages.

Keywords: Artificial Intelligence (AI); problem-solving skill; business innovation; qualitative research; case study; AI tools; data-driven insights; human–AI collaboration

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1. Introduction

Currently, the issues confronting managers are incredibly complex due to the surge of information, rapid technological advancements, and intensified global competition, which continue to exceed the capacity of traditional linear problem-solving techniques (Hisrich et al., 2017; Baron, 2006). Artificial intelligence (AI) is being utilized in this respect to support human judgment, automate repetitive decisions, and extract value from large

and rapidly evolving datasets (Davenport & Ronanki, 2018; Brynjolfsson & McAfee, 2014). Nevertheless, it has recently been observed that most companies are now applying AI to product development, sales, and marketing, making a transition towards data-driven decision-making (McKinsey & Company, 2021).

The early uses of AI in organizations were aimed at automating activities, though an increasing number of scholars are convinced that the more radical potential of AI is in the area of creativity and solving complicated problems by broadening the search space, generating alternatives, and challenging existing assumptions (Haase & Hanel, 2023; Binns & Veale, 2021). However, the effects of AI on creative thinking and teamwork in real-life organizations are hardly empirically evaluated, despite the rapid adoption of AI. The majority of available research is either conceptual (e.g., Shrestha et al., 2019) or descriptive, and there is a lack of comprehensive research on the micro-processes that drive the interactions between humans and AI when generating ideas, evaluating, and deciding on them in the workplace.

In theory, our study is based on the Componential Theory of Creativity, as suggested by Amabile, which posits that domain knowledge, creative processes, and motivation variables can be influenced by AI due to the accessibility of vast amounts of information and rapid iterations (Amabile, 1996, 2012). We also employ effectuation and design-thinking strategies that anticipate experimentation under uncertain conditions (Sarasvathy, 2001; Dew et al., 2009), as well as organizational conceptualizations of dynamic capabilities and decision routines when adapting to changing environments (Eisenhardt & Martin, 2000). These views are supplemented by the literature on the automation augmentation paradox, indicating that the balance of efficiency of algorithms and indefinable human feeling, contextual sense-making, and values has to be achieved to take advantage of AI (Raisch & Krakowski, 2021; Wilson & Daugherty, 2018).

Practically, startups and digital companies can be a central source of research on such dynamics due to their popularity and early adoption of machine learning, natural language processing, predictive analytics, and conversational agents. However, we lack in-depth insight into how these instruments redesign problem-solving orchestrations, influence creative ideation, and reorganize the division of cognitive labor between humans and AI (Westerman et al., 2022; Davenport & Ronanki, 2018).

In filling in these gaps, the current paper presents a qualitative single-case study of a medium-tech company that has introduced AI to several of its functions, including strategy, product, operations, marketing, sales, HR, analytics, and customer services. We examine the impact of AI on employees at various levels of the hierarchy in their day-to-day work and consider the implications of AI on organizational innovation. Accordingly, our research revolved around the following research questions:

- RQ1: How do organizations incorporate AI in their problem-solving strategies and decision-making?
- RQ2: How does AI involvement affect creative thinking and the development and optimization of strategic options?
- RQ3: In what ways can human AI collaborations, i.e., interaction between algorithmic cues and human judgment, facilitate or limit practical innovation?

The paper (i) offers empirical evidence on the cognitive and collaborative processes by which AI influences problem solving and creativity within a live business situation; (ii) synthesizes creativity, effectuation, and augmentation approaches to understand what was observed as such; (iii) makes practical suggestions as to how managers can pilot, scale, and govern AI to work alongside, instead of replacing, human ingenuity.

2. Literature Review

2.1. Redefining Problem-Solving in Modern Business

Business problem-solving depends entirely on skills, as these capabilities enable the identification of market problems and source recognition, leading to the development of suitable solutions for commercial use (Hisrich et al., 2017). The essential business capability in problem-solving provides innovation along with adaptability to businesses (Sarasvathy, 2001). Problem-solving expertise enables teams to navigate critical moments and devise innovative solutions that translate into business advantages, fostering successful competition (Baron, 2006).

Success for employees hinges on their ability to identify important opportunities and create effective solutions, according to Shane and Venkataraman (2000). Traditional linear business problem-solving methods cannot adapt to the current rapidly evolving business market environment. Startup businesses encounter complex problems that defy traditional definition because they require owners to create challenging yet necessary answers to vague business issues, according to Sarasvathy (2001). The conventional methods for problem solving demonstrate limited capability when dealing with adaptive system processes because they cannot efficiently manage multiple recursive variables. Design thinking and effectuation introduce novel theories that unite innovation and stakeholder engagement, while supporting experimental methods for better managing current problems (Dew et al., 2009).

2.2. Classical Problem-Solving Approaches

The classical approach to problem-solving has long been viewed as a progressive process, encompassing problem definition, information search, data analysis, solution generation, and implementation (Baron, 2006). This method is productive when problems are clear, but it breaks down when problems are ill-structured or complex and do not have clear solutions. Unpredictable and changing situations are especially faced by startups and entrepreneurial activities, which cannot be addressed with fixed planning (Sarasvathy, 2001).

To address these drawbacks, more recent models, such as design thinking and effectuation, focus on iteration, stakeholder involvement, and learning through trial and error. Design thinking proposes empathy and understanding of users, prototyping in a short duration, and refining it over time, whereas effectuation encourages entrepreneurs to do what they can with available resources and to develop solutions in collaboration with stakeholders (Dew et al., 2009). Collectively, these worldviews emphasize that contemporary problem-solving should be both analytical and exploratory, as well as creative.

2.3. AI Augmented Problem Solving

The incorporation of AI technologies into business processes is beginning to transform the way people approach problem-solving. The primary role of AI is to automate routine functions and process data at a volume and speed that people cannot manage (Brynjolfsson & McAfee, 2014). This can be applied in predictive analytics, which helps predict demand, automates customer support through chatbots, and classifies customers and optimizes marketing efforts through machine learning algorithms (Davenport & Ronanki, 2018; McKinsey & Company, 2021).

Nevertheless, researchers caution about the automation-augmentation paradox: in the case of AI, efficiency is improved, but excessive automation will lead to the loss of critical human judgment and situational awareness (Raisch & Krakowski, 2021). To reap the benefits of AI's full potential, organizations should utilize algorithmic insights to supplement human sensemaking and creativity, rather than perceiving AI as a potential labor-saving alternative.

2.4. Human–AI Co-Creation in Business and Design Contexts

A growing body of literature addresses collaborative intelligence, where human and AI systems work together to produce joint solutions. According to design thinking and innovation scenarios, AI serves as a creative partner, quickly generating prototypes, proposing new combinations of thought, and suggesting data-informed insights that disrupt existing assumptions (Westerman et al., 2022).

This type of human–AI collaboration enables organizations to combine the computational scale of AI with human intuition and ingenuity. However, most of the current research describes itself as abstract or technology-oriented, creating a gap in empirical data regarding the evolution of co-creation in the context of ordinary organizational practice.

Table 1 summarizes the selected literature to this study's theoretical and practical grounding linking AI to creativity, problem-solving and organizational decision-making.

Table 1. Summary of key literature on AI, creativity, and organizational decision-making.

Author(s), Year	Relevance to This Study
(Amabile, 2012)	Supports role of external stimuli (AI) in creativity
(Boden, 1998)	Used to frame how AI supports combinational creativity
(Shrestha et al., 2019)	Supports AI as complement to human problem-solving
(Jussupow et al., 2023)	Aligns with findings on operational efficiency
(Raisch & Krakowski, 2021)	Conceptually aligns with study's conclusion

2.5. Current Applications for Business Problem-Solving

The application of artificial intelligence (AI) continues to rise, improving problem-solving methods across various business operations. Machine learning-based predictive analytics enables businesses to foresee market trends, optimize supply chain operations, and improve customer group segmentation (McKinsey & Company, 2021). Clinical applications, such as IBM Watson, employ diagnostic algorithms that detect medical conditions while proposing therapeutic solutions, as AI functions as an assistant to human physicians (Topol, 2019). Similarly, AI-powered robotic systems and recommender modes accelerate business-to-business marketing activities by identifying promising leads and automating campaign personalization (Davenport & Ronanki, 2018).

However, the deployment and impact of AI systems differ widely depending on the organizational' s digital maturity and implementation strategy. Despite AI being known for its speed and efficiency, its role in enhancing creativity and innovation remains a contested issue. Most of the reported advantages, including cost savings or performance improvements, are dependent on perceptions or internal reporting by the participants, rather than independent, validated measurements, which indicate the need for a cautious approach to outcome interpretation.

Furthermore, scholars have observed that although AI may be useful in automating simple tasks, it often falls short in terms of complexity, providing a lack of context, and creative solutions to unstructured, complex problems (Brynjolfsson & McAfee, 2014). This has led business leaders to doubt that AI can create truly innovative solutions, rather than merely optimizing current processes. This has led to an increase in interest in hybrid methods that harness the efficiency of algorithms and the creative judgment and human understanding explored in detail in this study.

2.6. Theoretical Framework

The study builds on its previous points by applying theoretical frameworks to combine innovation, creativity, and problem-solving to maximize AI contributions. According to the Componential Theory of Creativity by Amabile (1996), researchers argue that AI stimulates creativity by enabling users to access advanced technologies and resources for coproduction and automation.

The implementation of machine learning algorithms enables support for both divergent and convergent thinking processes, allowing for the creation of innovative ideas and the identification of optimal solutions (Brynjolfsson & McAfee, 2014). The model employs Effectuation Theory, as proposed by Sarasvathy (2001), to illustrate creativity and innovation progress through the steps of hypothesis testing and feedback reception, leading to optimal solution adjustments.

Human proximity skills and intuitive capabilities remain beyond the current capabilities of AI to reproduce in basic operational tasks. The current challenge involves tackling intricate and disordered problems according to Davenport and Ronanki (2018).

As illustrated in Figure 1, the conceptual model displays the sequential relationship between artificial intelligence, human creativity, problem-solving skills, and business innovation, with the role of AI being to assist with the creative processes leading to the ensuing innovation. A theory-based hypothesis emerges from the findings that state:

- The understanding of knowledge, combined with the redundancy of effort, makes AI solutions possible for troubleshooting.
- The capabilities of AI enable employees to innovate through more effective convergence and divergence processes.
- The influence of AI on problem-solving and innovation among employees depends on how organizational cultures develop and are led by the management.

Figure 1. Visual conceptual model illustrating the directional relationship between Artificial Intelligence, Human Creativity, Problem-Solving Skills, and Business Innovation. Arrowheads indicate the flow of influence and progression among the components.

3. Materials and Methods

3.1. Research Design

The research investigates innovativeness, along with technological innovation potential, for AI technology adoption within an organization using a qualitative case methodology. Real-life case studies make the most appropriate research method because they enable a deep examination of actual situations (Yin, 2018). Given that the objective of this research was to understand the lived experiences of workers who work with AI tools, rather than testing predetermined hypotheses, a qualitative inquiry approach was the most suitable. In fact, the questions emphasize creativity and problem-solving in relation to AI, which justifies the use of qualitative methods instead of quantitative ones. These methods safeguard the complicated nature of multiple factors present in this study (Creswell & Poth, 2018).

3.1.1. Case Study Selection

The examined case presents a small to medium-tech start-up enterprise that applies Artificial Intelligence systems to both innovation and problem-solving operations. The organization represents an excellent choice because it contains detailed contextual information and intends to supply web solution services to external firms. The start-up extensively utilizes artificial intelligence resources, including machine learning tools and natural language processing tools,

to solve business problems and enhance its innovation capability (Brynjolfsson & McAfee, 2014). The startup was chosen because it was an early adopter of AI and had applied it to various areas within the organization. Being built with machine learning and NLP tools, it was a suitable choice for examining real-life uses of AI-powered solutions.

3.1.2. Sampling Strategy and Sampling Criteria

To select participants with the appropriate and sustained experience in using AI tools in their professional activities, a purposive sampling approach was employed (Ghezzi & Cavallo, 2020). The first step was to communicate with the company's leaders, who helped arrange meetings with employees in key departments. Diversity among the group's responsibilities, years of experience, and areas of work was prioritized to encompass a wide range of AI applications and perspectives.

- Participants were required to demonstrate at least two years of authentic work experience utilizing various AI tools, including predictive analytics, generative AI platforms, and automation systems, as part of their professional duties.
- The participants needed active involvement in AI-supported decision processes or problem-solving and innovative operations.
- Three organizational tiers within the company, ranging from founders to middle managers to frontline workers, participated to collect diverse insights about AI adoption and its organizational effects.
- The organization's reputation as an active adopter of AI provided a perfect setting.

Ten participants met the inclusion requirements, which included participation from three founders, four managers, and three operational personnel. The organization served as the perfect subject because it scaled AI-based services and expanded its operations worldwide (Jussupow et al., 2023).

The chosen participants satisfied this selection criterion, which enabled researchers to study practical AI implementations throughout different organizational functions and hierarchical levels, as follows in Table 2.

Table 2. Participants' criteria.

Participant ID	Role	Department/Function	Experience with AI Tools	Length of AI Use	Key AI Applications
P1	Founder	Strategy/Innovation	High	3 years	Business analytics, automation
P2	Founder	Product Development	High	2.5 years	Idea generation, prototyping
P3	Founder	Operations	Moderate	2 years	Workflow optimization, customer insights
P4	Middle Manager	Marketing	Moderate	2.5 years	Content generation, customer segmentation
P5	Middle Manager	Sales	Moderate	2 years	CRM tools, sales forecasting
P6	Middle Manager	HR/Talent	Moderate	2 years	Candidate screening, task automation
P7	Middle Manager	Data Analytics	High	3 years	Predictive analytics, dashboards
P8	Frontline Employee	Customer Support	Moderate	2 years	AI chatbots, customer query routing
P9	Frontline Employee	Marketing Support	Moderate	2 years	AI-assisted content suggestion
P10	Frontline Employee	Admin/Operations	Moderate	2 years	Document classification, workflow tracking

The study gained credibility through its established link to firsthand user experiences which conform to qualitative research principles when studying complicated organizational events (Raisch & Krakowski, 2021; Shrestha et al., 2019).

3.2. Data Collection

Primary data collection was conducted through semi-structured interviews, informed by the findings of the literature research. According to Creswell and Poth (2018), interviews proved to be a suitable research method because this technique enabled us to gather detailed observations and the interviewee's feelings and speculations. The interviews lasted between 45 and 60 min per participant and were conducted via Zoom software (Version: 6.5.12) for convenience. The interviews were recorded for easy data processing after obtaining consent.

The developed interview questions directly followed the framework established for research purposes. The respondents shared examples about AI's ease of problem-solving when compared to conventional methods as part of the evaluation of problem-solving achievement through AI tools (Davenport & Ronanki, 2018).

An interview protocol consisting of nine open-ended questions was developed and consisted of part of the interview method (see Appendix A). Questions are designed to address the study's three goals, highlighting AI's role in solving problems, fostering creativity, and collaborating with people. In interviews, participants spoke openly and shared many specific details about their experiences.

The assessment of creativity regarding AI included queries to determine if AI tools made it easier or harder for interviewees to create novel solutions (Amabile, 1996). The researchers asked participants to present ideas on how AI can help businesses develop creativity and innovation abilities (Brynjolfsson & McAfee, 2014). The data recordings and interviews took place on the Zoom platform before the transcripts were processed by Otter.ai. Reasons for analysis and information management involved assistance with the NVivo application.

3.3. Data Analysis

The methodology relied on thematic analysis, which followed Braun and Clarke's (2006, 2021) established six-phase protocol for qualitative data analysis. Multiple readings of interview transcripts served as the starting point to establish knowledge about the content.

Two researchers independently coded the transcripts to identify concepts that were found repeatedly. After that, the codes were brought together into main themes by comparing them over several iterations. NVivo was used as a tool to manage and arrange our codes. All themes were organized to ensure that the main ideas aligned with the study's objectives.

The data was manually coded to highlight important connections between artificial intelligence, creativity, and how people use their minds to solve problems. Different codes were grouped into broader categories based on their conceptual relationships with one another and their relevance to the research questions.

The analysis revealed three fundamental concepts about AI, including its use as a problem-solving tool and its ability to generate innovation, as well as the limitations inherent in AI-based creative approaches. Research findings were analyzed and evaluated using representative statements to both clarify and validate them through empirical evidence.

A requirement for moral research methods included obtaining informed consent from every participant prior to their data collection process. Participants received information about the study's purpose, methods, and their rights to remain anonymous and partici-

pate voluntarily. Transcripts contained no identifying data, while participants received anonymity protection through the transcription process.

The research employed the qualitative research standard of triangulation, comparing field observation notes with interview data, as described by Shrestha et al. (2019) and Jussupow et al. (2023). The single-case design provided suitable conditions for acquiring in-depth knowledge about a complex field, even though it produced findings with limited transferability to other situations.

3.4. Positionality and Bias Management

In this type of research, being aware of the researcher's background and working to eliminate bias ensures the trustworthiness of the findings. The study employed a "reflexive view" to help the research team avoid slipping into biases that might influence their interpretation of the findings.

Before the study began, the researchers recognized that artificial intelligence was becoming increasingly important in business settings. At the same time, efforts were made to avoid relying on past thoughts and to accept new themes that arose from their personal experiences. Using reflexive journaling, the researchers tracked and examined their own pre-existing opinions, decisions, and growth of understanding regarding the phenomena.

The data were collected through interviews that asked open-ended questions to ensure participants described their feelings without being influenced. To compare the various sources of data, interviews with participants were examined in conjunction with field notes and related literature to identify similarities and any differences. Although member checking could not be conducted, experts from our team checked and refined the codes to maintain the study's themes and consistency among members.

Because the study included contributions from people in various roles beyond management, it prevented any single perspective from dominating. To achieve an honest portrayal of AI-based innovation and creativity in organizations, this study took researcher influence into account and highlighted clear details about the methods used.

4. Results

A total of ten interview scripts obtained from the study served both as transcription material and summary content (Refer to Appendix A for the participants' respondents). Every participant in the interview sessions displayed the same responses when asked questions. Beneath the interviewing processes, AI systems acquire important knowledge that highlights their ability to drive business innovation through improved problem-solving. Through thematic analysis, three main patterns were identified that describe how startups adopt and utilize AI tools in their business operations listed in Table 3.

Table 3. Thematic dictionary.

Themes	Sub-Themes	Description	Illustrative Quotes
AI as a Strategic Problem-Solver	Automation, Speed, Data-Driven Decisions	Organizations gain operational speed through the integration of artificial intelligence while also achieving better business choices through this technology combination.	"Artificial intelligence tools enable us to handle information more efficiently thus generating quicker decisions."
AI as a Creativity Enabler	Brainstorming, Content Generation, Experimentation	AI enables idea generation through creative assistance which enriches human work without claiming their tasks.	"We use ChatGPT (GPT-4) to spark new product ideas or copy directions."

Table 3. *Cont.*

Themes	Sub-Themes	Description	Illustrative Quotes
Actionable Insights for Business	Collaboration, Upskilling, Pilot Testing	The implementation of AI adoption needs organizations to link it with employee training programs while it should scale up slowly.	"We started with a small AI use case before rolling it company-wide."

4.1. AI as a Strategic Problem-Solver

The respondents frequently mentioned AI as an effective tool that helps improve operational efficiency and reduce decision-making time.

Founders and managers emphasized perceived performance gains, with one founder (P1) reporting that "AI analytics tools reduced our supply chain costs by about 15%." These figures were not externally proven or validated; however, they reflect typical perceptions that AI-based analytics led to better resource allocation and uncovered inefficiencies that were previously undetected.

Several participants also noted that AI-powered chatbots managed a significant share of customer service interactions, with P2 estimating that "about 80% of our inquiries are now resolved automatically". Such a view is consistent with existing research indicating that the AI solutions are especially helpful in automating structured work, especially in repetitive jobs (Davenport & Ronanki, 2018).

From an analytical perspective, these findings suggest that AI primarily contributes to exploitative problem-solving (Eisenhardt & Martin, 2000), optimizing existing processes rather than generating radically new ones. The emphasis on speed and automation is consistent with Brynjolfsson and McAfee's (2014) view that AI excels at handling large-scale, data-intensive operations.

In the meantime, the subject participants understood limitations. Middle managers (P3, P4) stated that the issue is the inflated cost of implementation and maintenance. They had also pointed out that occasionally, AI was unable to read the contextual cues, and human oversight was required. This observation supports the automation-augmentation paradox (Raisch & Krakowski, 2021), underscoring that while AI enhances efficiency, its success depends on human supervision to ensure contextual relevance and accuracy.

4.2. AI as a Creativity Enabler

The respondents expressed the idea that AI tools inspire creativity, as they offer innovative ideas, data-driven information, and the ability to experiment quickly. The founders and marketing managers, specifically, claimed that AI assisted them in creating new product concepts, refining pricing solutions, and creating specific campaigns. In that regard, an example is provided by P5, who stated that AI-based price tracking boosted sales by approximately 10% which is an impression that proves that AI can offer practical market intelligence.

Analytically, such accounts posit that AI facilitates, firstly, combinational creativity (Boden, 1998) by assembling available data patterns to suggest new patterns or options. That was evident in the comment made by P4, who noted that AI prompts us to reconsider what we believed we knew and challenges our assumptions, a manifestation of reframing basic assumptions—a step towards breakthrough ideation (Amabile, 1996).

Nevertheless, the respondents were also wary of over-utilizing AI to generate ideas. P4 and P1 cautioned that over-automation may eliminate the diversity of ideas by too soon converging on data-driven solutions, which may be found to suppress divergent thinking. This contradiction highlights the augmentation issue: AI can be used to stimulate creativity,

but human judgment is necessary to shape and contextualize its recommendations (Raisch & Krakowski, 2021).

Combining these results, it can be concluded that the participants believe AI to be a useful partner in brainstorming and ideation, but it is not intended to replace human creativity. These findings should be interpreted with caution, as they are based on self-reports from participants and are limited to a particular case; however, they are indicative of AI creativity when combined with human judgment and review.

4.3. Actionable Insights for Businesses

Participants provided essential insights into how business organizations implement artificial intelligence systems in their operational activities. The deployment of a single AI project represents the highest potential for organizational success; following this, additional implementations should be initiated according to P1. According to participant P2, employee training constitutes the most crucial factor among all other elements. The expert recommends that business interventions should target distinct practical problems because "businesses should direct themselves toward real problems instead of merely gaining artificial intelligence systems."

The successful implementation of AI depends on achieving proper project goals using high-quality data, while maintaining regular interaction between domain experts and AI specialists. P1 and P2 declared that success factors create positive interactions between defined objectives and superior information systems alongside personnel with appropriate qualifications. Research participants identified human intuition, paired with AI analysis, as the optimal solution for addressing practical limitations related to AI use. The participants stated that organizations should combine AI analytical results with human intuition and practical expertise (P5) to enable workers to evaluate AI-generated outputs (P4).

5. Discussion

A thematic structural organization of all respondent data associated with the research question included references to relevant literature (see Table A1 in Appendix A). This research comprises three core aims: solving problems through testing AI software (ChatGPT-4), studying AI-based creativity, and delivering business-oriented analysis demonstrations for practitioners.

The article outlines how AI instruments help improve problem-solving abilities.

By implementing an AI analysis platform, the organization reduced expenses by 15% because it located supply chain inefficiencies (according to P1). Huge volumes of data become accessible to AI systems, which enables them to detect hidden patterns that human observers would normally not notice. Authors Brynjolfsson and McAfee (2014) demonstrate that AI produces greater value from complex data analysis than human operators.

Research statistics indicate that eighty percent of customer problems are successfully resolved by AI chatbots without human intervention, according to the information presented by P2. AI implements all techniques from educational content as defined by Davenport and Ronanki (2018). According to Agrawal et al. (2018), AI provides programs that utilize factual recommendations to enhance human decision-making capabilities.

The study participants explained specific limitations that appear when using AI systems to resolve issues. Analysis from P3 demonstrated that AI systems have intense maintenance requirements, while P4 highlighted the contextual problems that lead to errors within AI systems. Topol (2019) demonstrates through his report that machines will not exceed human-level performance when performing tasks executed at their peak capabilities. According to participant P5, the reliability of the statement "AI is dependent on good data" becomes questionable because it proves to be untrue at times. Amir et al. (2025) maintain a

position identical to that of Brynjolfsson and McAfee (2014), which states that the quality of training data collections has a significant impact on AI performance.

AI deployment in problem-solving practices made substantial advancements visible to survey participants despite their understanding of system operational boundaries. P1 noted how AI performs fast processing of enormous datasets that human operators would never complete. The statement made by P2 clarified that "AI launches distinct decision-making patterns because it supplies analytical and instinctive processing capabilities." AI systems make objective business decisions based on findings that align with the work of Davenport and Ronanki (2018). Shrestha et al. (2019) developed research that shows AI analyzing massive live data to help organizations better track market developments.

Results-Oriented Business Solutions

Stakeholders proposed solutions for businesses seeking to participate in AI contracts, which provided them with potential business opportunities. The first expert advocated beginning experimental projects to discover AI strengths before extensive implementation (P1), and the second expert stated that staff training must receive funding so that employees become proficient with AI tools (P2). Davenport and Ronanki (2018) support phased AI implementation through incremental approaches, accompanied by ongoing staff training based on research findings.

The publication by Fountaine et al. (2019) recommends launching minimal test programs to achieve peak organizational readiness through pilot programs. The most important components for successful AI implementation include defining specific objectives and using high-quality information along with a coordinated relationship between subject matter experts and AI technical specialists. P1 identified "good data together with clear goals as well as capable people" as the key to success, while P2 emphasized the importance of maintaining "the right ratio between field professionals and AI specialists." The two perspectives highlight the importance of aligning AI project objectives with organizational goals to ensure effective collaboration (Brynjolfsson & McAfee, 2014).

According to Wilson and Daugherty (2018), the effective use of augmented AI relies on collaboration between human resources and machines. Research participants argued that organizations should implement both human expert perspectives along with artificial intelligence knowledge to produce optimal results, and P4 emphasized the importance of human stakeholders in reviewing artificial intelligence suggestions. The adoption of AI requires human involvement, according to numerous sources (Topol, 2019) to maintain accurate and current outputs. Raisch and Krakowski (2021) establish that recognizing human–AI interaction as a fundamental risk variable in AI practices is crucial for users.

6. Conclusions

6.1. Key Findings

This paper focused on understanding two pivotal questions regarding artificial intelligence use in organizations: (1) What methods do Organizations use to assimilate AI into their problem-solving operations, and (2) Does AI impact creativity during workflow innovation?

Employees at the site start-up highlighted that AI systems enhance organizational problem-solving through task automation and workflow optimization, as well as automated strategic insights from processed data, according to qualitative research. Studies about AI-enhanced decision-making under pressure by Jussupow et al. (2023) and Shrestha et al. (2019) mirror the current research findings.

The study identified specific methods by which AI technology facilitates creative output in response to the second research question. Study participants indicated that

AI mechanisms improve creativity by providing statistical data insights as well as mind-altering mental perspectives and digital-based experimental creative opportunities.

6.2. Theoretical Contribution and Practical Implications

The study results support the predictions of Amabile's (2012) Componential Theory of Creativity regarding the positive impact of accessing domain-relevant information and support for the creativity process on creative work. Artificial intelligence systems lack two vital elements of creative originality: intrinsic motivation and contextual sensitivity, which is why human judgment and vision continue to be necessary.

The findings are contextualized within Boden's (1998) creative framework, which differentiates between three categories of creativity: combinational, exploratory, and transformational. Currently deployed artificial intelligence systems utilize combinational creativity features, which involve creating new combinations from already existing ideas or data structures. Modern humans maintain exclusive control over revolutionary creative approaches through their inherent intuition while merging it with emotional insights.

The study can be considered useful for business professionals, as it offers specific guidelines that they can put into action. In fact, business leaders could:

- Implement AI systems as co-collaborators for projects that blend structured tasks with innovative thinking, like content production, design thinking, and marketing strategy development.
- Establish broad internal competence in AI, which allows personnel to harness these systems with both a comprehensive understanding and original thinking.
- Scale AI implementation begins in non-critical creative areas before progressing to vital processes following thorough quality checks.

The study provides companies with practical knowledge that demonstrates why they should exercise caution when trusting AI systems to solve complex, contextual problems. The study participants stated that future AI adoption depends on developing skills among the team and establishing better partnerships between artists, designers, and AI specialists. The findings align with Davenport and Ronanki's (2018) position that AI deployment is correlated with business support, as well as cultural coherence. Strategic human-centered methods that combine AI technology enhancement with human imagination have led to effective ways of overcoming AI limitations and generating lasting innovations.

Findings from this case suggest that AI systems, when combined with human input, can spark creative insights. Nonetheless, more research is needed to confirm this effect across various contexts.

6.3. Challenges, Limitations, and Future Work

This paper has several shortcomings, despite its findings. The study relied on a single-case analysis, limiting the ability to generalize results across different industries. Additional research is needed on multi-case examinations of various sectors involving different organizational entities.

This study relied on interviews with only ten people from a single organization, which may limit the applicability of its findings to other groups. Accordingly, results should be interpreted as context-specific insights rather than generalizable conclusions; further research should employ multi-case designs and larger samples to achieve broader applicability. In addition, although selecting participants with genuine experience helped ensure accuracy, collecting such data through self-report can introduce biases. For this reason, future research should employ multiple methods to establish the study's reliability, such as matching interview data with our notes, having different researchers code the findings, and reflecting on how the study was conducted. While the results were shaped

by the company's unique conditions, they have useful lessons for similar tech-oriented companies working with AI. Future research should investigate more detailed cases and employ various research strategies to compare views and confirm the reported results.

6.4. Closing Reflection

The research demonstrates a fundamental understanding of academic and practice, suggesting that artificial intelligence could stimulate human mental growth rather than merely completing automated processes. Creativity is a fundamental human practice that develops through environmental, emotional, and semiotic factors. The purposeful application of AI technology becomes a tool that lifts organizational capabilities without the intention of replacement so organizations can discover fresh innovative dimensions. The intelligent business of the future requires a partnership between humans and technology to develop solutions that exceed the individual capabilities of people or computers.

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Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
GPT	Generative Pre-Trained Transfer
P	Participant

Appendix A

Table A1. Participants Responses to Interview Questions.

Question 1: Can You Describe a Specific Instance Where AI Tools Helped You Solve a Business Problem?	Question 2: How Has AI Changed Your Approach to Problem-Solving Compared to Traditional Methods?	Question 3: What Challenges Have You Encountered When Using AI for Problem-Solving?
"We used an AI-powered analytics tool to identify inefficiencies in our supply chain, reducing costs by 15%."	"AI allows us to process large datasets quickly, which wasn't possible with manual methods."	"AI sometimes provides irrelevant insights, requiring us to manually filter the data."
"AI chatbots helped us resolve 80% of customer queries without human intervention."	"AI has made problem-solving more data-driven and less reliant on intuition."	"Training the AI models requires significant time and expertise."
"We used machine learning to predict customer churn and implemented targeted retention strategies."	"AI has enabled us to test multiple solutions simultaneously, speeding up decision-making."	"AI tools can be expensive to implement and maintain."

Table A1. *Cont.*

Question 1: Can You Describe a Specific Instance Where AI Tools Helped You Solve a Business Problem?	Question 2: How Has AI Changed Your Approach to Problem-Solving Compared to Traditional Methods?	Question 3: What Challenges Have You Encountered When Using AI for Problem-Solving?
"AI-driven sentiment analysis helped us understand customer feedback and improve our product."	"AI has shifted our focus from reactive to proactive problem-solving."	"AI lacks contextual understanding, leading to occasional misinterpretations."
"We used AI to optimize our marketing campaigns, increasing ROI by 20%."	"AI has reduced human bias in decision-making by providing objective insights."	"AI requires high-quality data, which isn't always available."
Question 4: How do AI tools influence your creative process when developing new solutions?	Question 5: Can you provide an example of a creative idea that emerged from using AI?	Question 6: Do you think AI enhances or limits creativity? Why?
"AI tools provide data-driven insights that inspire new ideas we wouldn't have considered otherwise."	"We used AI to generate personalized product recommendations, which increased customer engagement."	"AI enhances creativity by offering new perspectives, but it can't replace human intuition."
"AI helps us brainstorm by generating multiple options based on data patterns."	"AI suggested a new pricing strategy that boosted sales by 10%."	"AI enhances creativity but requires human oversight to refine ideas."
"AI tools allow us to experiment with creative solutions in a risk-free virtual environment."	"We used AI to design a new user interface that improved customer satisfaction."	"AI enhances creativity by automating repetitive tasks, freeing up time for innovation."
"AI provides unconventional insights that challenge our assumptions and spark creativity."	"AI helped us create a dynamic content strategy that adapts to real-time trends."	"AI enhances creativity but can limit it if over-relied upon for idea generation."
"AI tools help us identify gaps in the market, which we use to develop innovative products."	"We used AI to develop a gamified loyalty program that increased customer retention."	"AI enhances creativity by providing data-driven inspiration, but human input is crucial."
Question 7: What advice would you give to other businesses considering AI adoption for problem-solving and creativity?	Question 8: What factors contribute to the successful integration of AI into problem-solving processes?	Question 9: How can businesses overcome the limitations of AI in fostering creativity?
"Start small with pilot projects to understand AI's potential before scaling up."	"Clear objectives, quality data, and skilled personnel are critical for success."	"Combine AI insights with human creativity to overcome its limitations."
"Invest in training your team to use AI tools effectively."	"Collaboration between AI experts and domain specialists is essential."	"Use AI as a tool to augment, not replace, human creativity."
"Focus on solving specific problems rather than adopting AI for the sake of it."	"A culture of experimentation and openness to failure is key."	"Regularly review and refine AI outputs to ensure relevance and accuracy."
"Choose AI tools that align with your business goals and capabilities."	"Leadership support and a clear implementation strategy are crucial."	"Encourage employees to think critically about AI-generated ideas."
"Partner with AI vendors who offer robust support and training."	"Continuous monitoring and feedback loops ensure AI remains effective."	"Balance AI-driven insights with human intuition and experience."

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